

HW0505

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11.24

a.

Table: New Reduced 'P' equation

term	estimate	std.error	statistic	p.value
(Intercept)	-0.3596	0.0792	-4.5426	0.0000
mon	-0.1082	0.1034	-1.0470	0.2975
tue	-0.0661	0.1010	-0.6542	0.5145
wed	-0.0493	0.1038	-0.4748	0.6359
thu	0.0393	0.1007	0.3907	0.6968
stormy	0.4463	0.0792	5.6348	0.0000
mixed	0.2369	0.0785	3.0170	0.0032

在 5% 的顯著水準下，Mixed 變數為顯著 ($p\text{-value} = 0.0032 < 0.05$)。

Analysis of Variance Table

Model 1: $\text{lprice} \sim \text{mon} + \text{tue} + \text{wed} + \text{thu}$

Model 2: $\text{lprice} \sim \text{mon} + \text{tue} + \text{wed} + \text{thu} + \text{stormy} + \text{mixed}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F 檢定值為 $16.14 > 10$ 。

b.

2SLS estimates for 'Demand' (equation 1)

Model Formula: `lquan ~ lprice + mon + tue + wed + thu`

Instruments: `~mon + tue + wed + thu + stormy + mixed`

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.5402344	0.1566081	54.53250	< 2.22e-16 ***
lprice	-0.9301409	0.3533995	-2.63198	0.0097682 **
mon	-0.0119020	0.2083710	-0.05712	0.9545586
tue	-0.5258315	0.2022954	-2.59932	0.0106845 *
wed	-0.5626197	0.2069581	-2.71852	0.0076735 **
thu	0.0998706	0.2028454	0.49235	0.6235011

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.685304 on 105 degrees of freedom

Number of observations: 111 Degrees of Freedom: 105

SSR: 49.312422 MSE: 0.469642 Root MSE: 0.685304

Multiple R-Squared: 0.185033 Adjusted R-Squared: 0.146225

具有顯著性的變數維持不變。不過其係數估計值的絕對值，經過加入 STORMY 和 MIXED 作為工具變數跑 2SLS 後，都有縮小（特別是 $\ln(\text{Price})$ 的部分）。

c.

Sargan Test Statistic: 0.7721695

```
> cat("P-value:", p_val, "\n")
```

P-value: 0.3795467

p-value = 0.3795467，明顯無法拒絕虛無假說（工具變數是有效的）。

d.

Analysis of Variance Table

Model 1: $\text{lprice} \sim \text{stormy} + \text{mixed}$

Model 2: $\text{lprice} \sim \text{mon} + \text{tue} + \text{wed} + \text{thu} + \text{stormy} + \text{mixed}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

p-value = 0.6437，星期幾作為工具變數不具聯合顯著性，會使第二階段出現共線性問題，We are not able to estimate the supply equation by 2SLS with confidence.

11.28

a.

原本的方程式 (Q 在左邊) :

- Demand: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 P S_i + \alpha_4 D I_i + e_{di}$
- Supply: $Q_i = \beta_1 + \beta_2 P_i + \beta_3 P F_i + e_{si}$

我們將 P_i 移到等號左邊，其他移到右邊，得到反需求方程式與反供給方程式：

- 反需求 (Inverse Demand):

$$P_i = -\frac{\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} Q_i - \frac{\alpha_3}{\alpha_2} P S_i - \frac{\alpha_4}{\alpha_2} D I_i - \frac{1}{\alpha_2} e_{di}$$

令新參數為 γ :

$$P_i = \gamma_1 + \gamma_2 Q_i + \gamma_3 P S_i + \gamma_4 D I_i + u_{di}$$

- 反供給 (Inverse Supply):

$$P_i = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} Q_i - \frac{\beta_3}{\beta_2} P F_i - \frac{1}{\beta_2} e_{si}$$

令新參數為 θ :

$$P_i = \theta_1 + \theta_2 Q_i + \theta_3 P F_i + u_{si}$$

預期符號 (Anticipated Signs) :

根據經濟學直覺與課本 Example 11.1 的設定：

1. 需求法則：原 $\alpha_2 < 0$ ，因此 $\gamma_2 = 1/\alpha_2 < 0$ (需求曲線為負斜率)。
2. 替代品價格 (PS)：原 $\alpha_3 > 0$ ，因此 $\gamma_3 = -\alpha_3/\alpha_2 > 0$ 。
3. 可支配收入 (DI)：假設為正常財，原 $\alpha_4 > 0$ ，因此 $\gamma_4 = -\alpha_4/\alpha_2 > 0$ 。
4. 供給法則：原 $\beta_2 > 0$ ，因此 $\theta_2 = 1/\beta_2 > 0$ (供給曲線為正斜率)。
5. 生產要素成本 (PF)：成本上升使供給減少，原 $\beta_3 < 0$ ，因此 $\theta_3 = -\beta_3/\beta_2 > 0$ 。

b.

2SLS estimates for 'InvDemand' (equation 1)

Model Formula: $p \sim q + ps + di$

Instruments: $\sim ps + di + pf$

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.42841	13.59161	-0.84084	0.4081026
q	-2.67052	1.17495	-2.27287	0.0315350 *
ps	3.46108	1.11557	3.10252	0.0045822 **
di	13.38992	2.74671	4.87490	4.6752e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.165551 on 26 degrees of freedom

Number of observations: 30 Degrees of Freedom: 26

SSR: 4506.625289 MSE: 173.331742 Root MSE: 13.165551

Multiple R-Squared: 0.556717 Adjusted R-Squared: 0.505569

2SLS estimates for 'InvSupply' (equation 2)

Model Formula: $p \sim q + pf$

Instruments: $\sim ps + di + pf$

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-58.798223	5.859161	-10.0353	1.3165e-10 ***
q	2.936711	0.215772	13.6103	1.3212e-13 ***
pf	2.958486	0.155964	18.9690	< 2.22e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399078 on 27 degrees of freedom

Number of observations: 30 Degrees of Freedom: 27

SSR: 522.500877 MSE: 19.351884 Root MSE: 4.399078

Multiple R-Squared: 0.948605 Adjusted R-Squared: 0.944798

Signs are all correct.全部係數皆顯著異於零。

c.

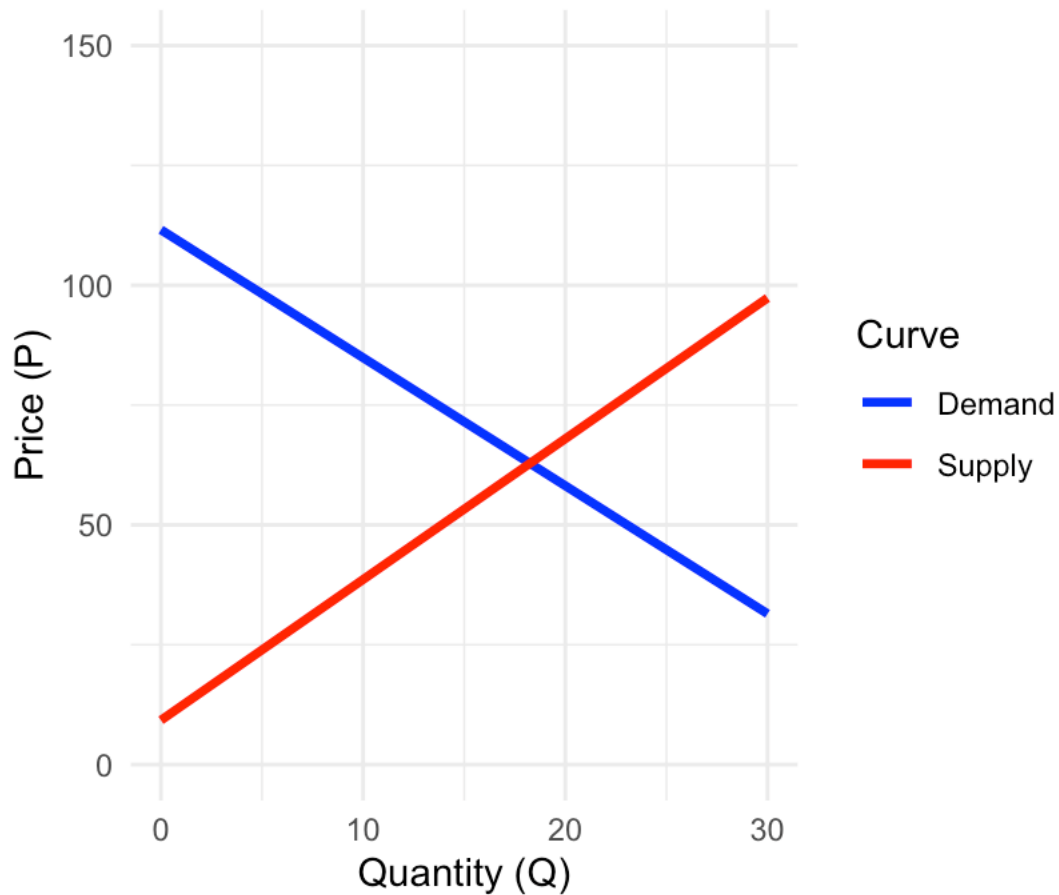
Price Elasticity of Demand at the means: -1.272464

d.

```
> cat("Demand Equation for Sketch: P =", int_D, "+ (", slope_D, ") * Q \n")
Demand Equation for Sketch: P = 111.5801 + ( -2.670519 ) * Q
> cat("Supply Equation for Sketch: P =", int_S, "+ (", slope_S, ") * Q \n")
Supply Equation for Sketch: P = 9.246963 + ( 2.936711 ) * Q
```

Truffle Market: Supply and Demand Curves

Given: DI = 3.5 , PS = 22 , PF = 23



c.

```
> cat("=== 方法一：結構方程式 (2SLS) 解出之均衡點 ===\n")
=== 方法一：結構方程式 (2SLS) 解出之均衡點 ===
> cat("預測均衡數量 Q* =", q_star_struct, "\n")
預測均衡數量 Q* = 18.25021
> cat("預測均衡價格 P* =", p_star_struct, "\n\n")
預測均衡價格 P* = 62.84257

> cat("=== 方法二：縮減式 (Reduced-form) 預測之均衡點 ===\n")
=== 方法二：縮減式 (Reduced-form) 預測之均衡點 ===
> cat("預測均衡數量 Q* =", q_star_red, "\n")
預測均衡數量 Q* = 18.2604
> cat("預測均衡價格 P* =", p_star_red, "\n")
預測均衡價格 P* = 62.81537
```

基本上兩者方式算出的 equilibrium values 答案會大致相同，若有也僅是小數點後誤差。

f.

Call:

```
lm(formula = inv_D, data = truffles)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-25.0753	-2.7742	-0.4097	4.7079	17.4979

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-13.6195	9.0872	-1.499	0.1460
q	0.1512	0.4988	0.303	0.7642
ps	1.3607	0.5940	2.291	0.0303 *
di	12.3582	1.8254	6.770	3.48e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom

Multiple R-squared: 0.8013, Adjusted R-squared: 0.7784

F-statistic: 34.95 on 3 and 26 DF, p-value: 2.842e-09

Call:

```
lm(formula = inv_S, data = truffles)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-8.4721	-3.3287	0.1861	2.0785	10.7513

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-52.8763	5.0238	-10.53	4.68e-11 ***
q	2.6613	0.1712	15.54	5.42e-15 ***
pf	2.9217	0.1482	19.71	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom

Multiple R-squared: 0.9531, Adjusted R-squared: 0.9496

F-statistic: 274.4 on 2 and 27 DF, p-value: < 2.2e-16

Demand equation 在 OLS 不僅 q 的 sign 不是負向，且也是唯一一個變不顯著。